SONG YUEXI

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Portfolio | GitHub | LinkedIn



Research Interests

Autonomous underwater vehicles, robot learning and control, safe and robust embodied AI, computer vision, and sports analytics.

Education

National University of Singapore Aug 2023–Jun 2027
B.Comp. in Computer Science GPA: 4.71/5.0
Specializations in AI and Security; Second Major in Mathematics Dean's List (Top 5%)

Georgia Institute of Technology Jan 2026–May 2026
Semester Exchange GPA: 4.0/4.0

Yale University May 2024–Jul 2024
Summer Session GPA: 4.0/4.0

Publications

BASeg: Boundary-Aware Remote Sensing Segmentation with Structural Penalties [Paper]
Yuexi Song, Kailai Sun, Mingyi He, Paul Pu Liang, Shenhao Wang, and Jinhua Zhao
WACV 2026 @ CV4EO; Under review at AAAI 2027.

Robustness without Wrinkles: Parallel Simulation and Robust MPC for Certified Deformable Manipulation [Paper]
Wei-Chen Li*, Jeffrey Fang*, Sasanka Polisetti, **Yuexi Song**, and Glen Chou
CoRL 2026; RSS 2026 @ Beyond Regidity.

TANS: A Chess-Inspired Notation System for Strategy Analysis of Tennis Games [Paper]
Yuexi Song, Chuanfei Li, Hao Cao, Ling Wu, Huanhuan Zheng, and Zhenkai Liang
International Sports Analytics Conference and Exhibition 2025. Distinguished Paper Award (Top 5%).

TANS-Agent: Structured, Multi-agent Tactical Reasoning for Tennis Match Analysis and Strategy Recommendation [Paper]
Emma Y. Cai, **Yuexi Song**, Chuanfei Li, and Zhenkai Liang
International Sports Analytics Conference and Exhibition 2026.

Research & Robotics Experience

Robotics Research Assistant Aug 2026–Present
CLeAR Lab, National University of Singapore (Advisor: Prof. Harold Soh)

- Formulating diagnostic frameworks to interpret failure modes in embodied AI foundation models under environment distribution shifts as part of Final Year Project.

Robotics Research Assistant Dec 2025–Present
Trustworthy Robotics Lab, Georgia Institute of Technology (Advisor: Prof. Glen Chou)

- Formulated and deployed MuJoCo simulation environments for complex deformable object manipulation, enabling rigorous safety and control benchmark evaluations.
- Adapted parallelized GPU system-level synthesis (SLS) solvers to achieve real-time certified trajectory generation for deformable cable manipulation.
- Investigated LQR-guided activation steering for Vision Transformers to mitigate feature degradation under visual domain shifts.

Computer Vision Research Intern May 2025–Aug 2025
Singapore–MIT Alliance for Research and Technology (Advisor: Prof. Jinhua Zhao)

- Proposed MABL, a plug-in loss using Mahalanobis-distance weighting to emphasize structurally important pixels and angle-aware penalties to align predicted boundary orientations.
- Designed and implemented BASeg, a semantic segmentation model integrating Vision Mamba and spatial convolution layers; ran multi-GPU evaluation across four datasets, improving mIoU by up to 2.8%.

Robotics Research Assistant May 2025–Dec 2025
Human-Centered Robotics Lab, National University of Singapore (Advisor: Prof. Shi Fan)

- Constructed a 10,000-instance PDDL benchmark for safety-aware mission planning in autonomous systems.
- Built an LLM-driven pipeline to generate high-coverage, safety-critical planning scenarios.

Honors, Fellowships & Awards

NUS Research Experience (REx) Fellow, National University of Singapore
NUS Robotics Achievement Award, National University of Singapore
1st Place, Autonomy Challenge, RoboSub
Honor List of Student Tutors, National University of Singapore
3rd Place, GIC Code to Impact Hackathon
NUS Science & Technology Undergraduate Scholarship, National University of Singapore
2nd Prize, China National Physics Olympiad, Shandong Regional

Teaching Experience

Teaching Assistant Aug 2024–Present
National University of Singapore Singapore

- Hosted weekly tutorial sessions of 20 students and mass consultations before midterms and finals.
- Courses taught: CS1231S Discrete Structures; CS2040S Data Structures and Algorithms; CS3230 Design and Analysis of Algorithms.

Industry Experience

Software Engineer May 2026–Aug 2026
Goldman Sachs Singapore

- Developed an AI-powered agentic workflow system to streamline automated decision-making and risk-assessment pipelines for risk managers.
- Engineered cloud-native production migrations on AWS, integrating Snowflake data workflows, custom user-defined table functions, and robust end-to-end deployment validation.

Software Engineer Sep 2023–Dec 2025
Bumblebee Autonomous Systems Singapore

- Architected a real-time YOLO-based perception pipeline for underwater object detection on autonomous underwater vehicles.
- Developed mission-planning algorithms to support autonomous task execution.
- Integrated perception, navigation, control, and hardware subsystems through ROS2, contributing to a 1st-place finish in the RoboSub 2025 Autonomy Challenge.

Risk Advisory Intern Dec 2024–Jan 2025
Deloitte Singapore

- Conducted systematic developer testing for an enterprise secure large-file transfer system.
- Validated cryptographic encryption and decryption correctness across end-to-end data transfer workflows.

Projects

Tennis Algebraic Notation System (TANS) – Tennis Analytics Platform Jan 2025–Present
Developed a formal notation for data-driven tennis strategy analysis and applied grey relational analysis to identify winning patterns. [Project]

TennisPro – iOS Application May 2024–Aug 2024
Built a SwiftUI application with Firebase and ML Kit for tennis movement analysis, authentication, and data extraction.

Hornet – Autonomous Underwater Vehicle Sep 2023–May 2024
Developed CAN-bus device communication with a 30-member team for SAUVC, placing among the top 15 teams.

Leadership Experience

Team Manager, NUS Varsity Tennis Team Apr 2025–Dec 2025
Coordinate weekly training plans and lead recruitment and onboarding.

Project Director, BizCom League October 2023–Mar 2024
Led a 10-member team to design and organize a two-day NUS Students' Computing Club event for 50 participants.

Technical Skills

Programming Languages: Python, Java, C++, C, SQL, Swift, JavaScript, HTML/CSS
Machine Learning & Vision: PyTorch, OpenCV, Hugging Face
Robotics & Simulation: ROS2, MuJoCo, Isaac Sim
Systems & Infrastructure: Linux/Ubuntu, Bash, Docker, Git, AWS, Flask, Firebase, Xcode, Wireshark